

Types and variables

Intro Java 1

Clicker Questions

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How long does the reading take?

- A. 1 hours
- B. 2 hours
- C. 3 hours
- D. 4 hours
- E. 5+ hours

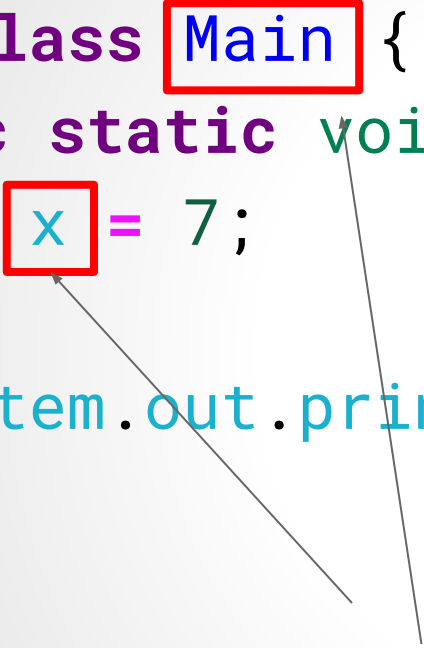
Identifiers in Java

An *identifier* is the name of something, such as a variable or the name of the program

Identifiers must start with a letter or underscore, and then contain 0 or more letters, digits, underscores

Identifiers in Java

```
public class Main {  
    public static void main(String[] args) {  
        int x = 7;  
  
        System.out.println("x is " + x);  
    }  
}
```

A diagram with two red rectangular boxes. The first box is around the word 'Main' in the line 'public class Main {'. The second box is around the variable 'x' in the line 'int x = 7;'. Two thin black arrows originate from the bottom-right corner of the first box and the bottom-left corner of the second box, both pointing towards the text 'These are identifiers' located at the bottom right of the slide.

These are identifiers

Which of these are legal Java identifiers?

- A. `5int`
- B. `the int`
- C. `int`
- D. all of the above
- E. none of the above

Which of these are legal Java identifiers?

- A. 5int
- B. the int
- C. int
- D. all of the above
- E. none of the above

int is not a legal identifier because int already has a special meaning in Java---it means a variable of type int.

These are called *reserved words* or *keywords*, and you can't use them for variable names.

Is this legal Java syntax?

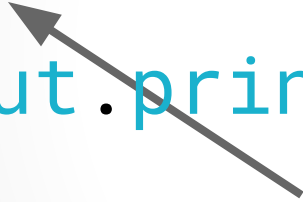
```
int x=5;  
System.out.println("x is "+x);  
int x=7;  
System.out.println("now x is "+x);
```

A: yes!

not A: no!

Is this legal Java syntax?

```
int x=5;  
System.out.println("x is "+x);  
int x=7;  
System.out.println("now x is "+x);
```



A: yes!

not A: no!

Once you create a variable named x, you cannot re-create another variable named x in the same scope (i.e. inside the same group of curly braces, omitted here but presumed; right now we only have one scope for all of our code so this is easy to track).

So this will not work!

<https://repl.it/@jspacco/141-x-is-7>

Is this legal Java syntax?

```
int x=5;  
System.out.println("x is "+x);  
x=7;  
System.out.println("now x is "+x);
```

A: yes!

not A: no!

Is this legal Java syntax?

```
int x=5;  
System.out.println("x is "+x);  
x=7;  
System.out.println("now x is "+x);
```

A: yes!

Yes, this is totally fine!

not A: no!

x is created with a value of 5, then given the new value of 7. Totally legal and will work correctly.

Is this legal Java syntax?

```
double d = 100;
```

A: yes

not A: no

Is this legal Java syntax?

```
double d = 100;
```

A: yes

not A: no

Java can always convert integers to doubles by adding a decimal place.

This is usually called *type promotion* or *type conversion*.

Java is happy to convert, so long as there will be no loss of precision.

Will this work?

```
int x = 0.0;
```

A: yes!

not A: no!

Will this work?

```
int x=0.0;
```

This will not work, because the types are not compatible.

A: yes!

0.0 is a *double*, while x is declared as an *int*.

not A: no!

Java is maximally conservative, and will never automatically convert doubles to ints.

Humans can see that no rounding is involved, but computers are not that smart...

The fear is *loss of precision*, i.e. rounding.

What does this print?

```
double ftemp = 75;  
double ctemp = (ftemp-32) * (5/9);  
System.out.println(ctemp);
```

- A. 0.0
- B. 23.0
- C. 23.88888888888889
- D. it won't compile
- E. none of the above

What does this print?

```
double ftemp = 75;  
double ctemp = (ftemp-32) * (5/9);  
System.out.println(ctemp);
```



A. 0.0

Argh, integer division!

B. 23.0

5/9 is 0!

C. 23.88888888888889

Anything * 0 is 0!

D. it won't compile

E. none of the above

How many of these correct the previous problems?

- `double ctemp = ftemp - 32 * 5 / 9.0;`
- `double ctemp = (ftemp - 32) * (5.0 / 9);`
- `double ctemp = (ftemp-32) * 5.0 / 9.0;`
- `double ctemp = 5 * (ftemp-32) / 9;`

- A. 0
- B. 1
- C. 2
- D. 3
- E. 4

remember that `ftemp` is a double

How many of these correct the previous problems?

- `double ctemp = ftemp - 32 * 5 / 9.0;`
- `double ctemp = (ftemp - 32) * (5.0 / 9);`
- `double ctemp = (ftemp-32) * 5.0 / 9.0;`
- `double ctemp = 5 * (ftemp-32) / 9;`

- A. 0
- B. 1
- C. 2
- D. 3**
- E. 4

How many of these correct the previous problems?

- `double ctemp = ftemp - 32 * 5 / 9.0;`
- `double ctemp = (ftemp - 32) * (5.0 / 9);`
- `double ctemp = (ftemp-32) * 5.0 / 9.0;`
- `double ctemp = (ftemp-32) * 5 / 9.0;`

Since temp is double, this also works:

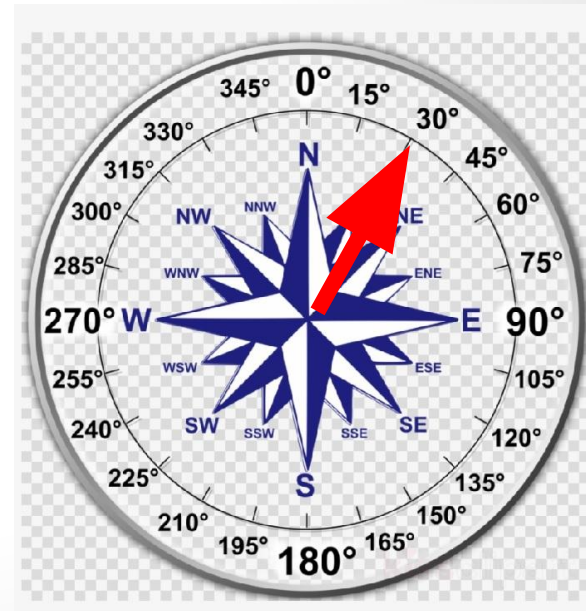
```
double ctemp = (ftemp-32) * 5 / 9;
```

- A. 0
- B. 1
- C. 2
- D. 3**
- E. 4

**Suppose we are heading 30 degrees
and we turn 400 degrees to the right**

What is our new heading?

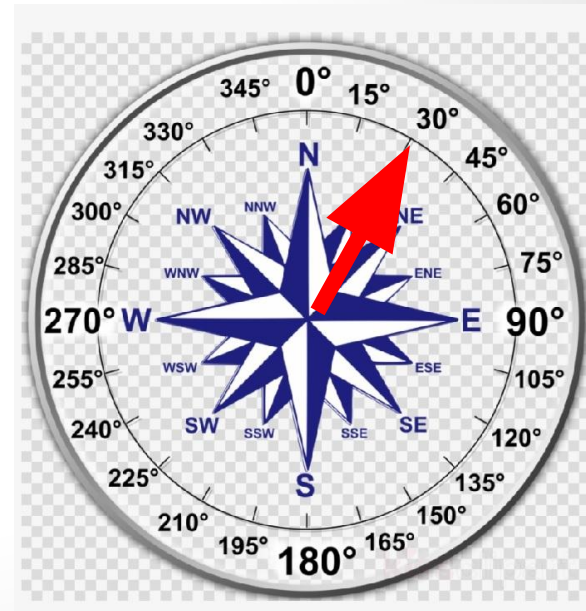
- A. 30 degrees
- B. 430 degrees
- C. 70 degrees
- D. 90 degrees
- E. I'm not lost! I don't
need directions!



Suppose we are heading 30 degrees and we turn 400 degrees to the right

What is our new heading?

- A. 30 degrees
- B. 430 degrees
- C. 70 degrees**
- D. 90 degrees
- E. I'm not lost! I don't need directions!



Suppose we are heading D degrees and we turn right N degrees

What is our new heading? Assume D is
between 0 and 359 and D is positive

- A. $D + N$
- B. $D + N / 360$
- C. $(D + N) / 360$
- D. $D + N \% 360$
- E. $(D + N) \% 360$

Suppose we are heading D degrees and we turn right N degrees

What is our new heading? Assume D is
between 0 and 359 and D is positive

- A. $D + N$
- B. $D + N / 360$
- C. $(D + N) / 360$
- D. $D + N \% 360$
- E. $(D + N) \% 360$

Let's do a couple of examples on
the board and see why

Suppose we are heading D degrees and we turn right N degrees

What is our new heading? Assume D is
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- A. $D + N$
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- C. $(D + N) / 360$
- D. $D + N \% 360$
- E. $(D + N) \% 360$

Let's do a couple of examples on
the board and see why

**DONE FOR NOW
MOVING ON TO LOOPS**

Extra slides follow

Suppose you have ponies behind 6 doors numbered 0 1 2 3 4 5



0

1

2

3

4

5

You want to shift each pony 3 spots to the right, wrapping back around if necessary.

Suppose you have ponies behind 6 doors numbered 0 1 2 3 4 5



0

1

2

3

4

5

You want to shift each pony 3 spots to the right, wrapping back around if necessary.

For example, to shift Applejack (door 5) right 1 puts them into door 0 (where Twilight Sparkle is currently located)

Suppose you have ponies behind 6 doors numbered 0 1 2 3 4 5



0

1

2

3

4

5

You want to shift each pony 3 spots to the right
Where would Twilight Sparkle (door 0) end up?

- A. 1
- B. 2
- C. 3
- D. 4
- E. 5

Suppose you have ponies behind 6 doors numbered 0 1 2 3 4 5



You want to shift each pony 3 spots to the right
Where would Twilight Sparkle (door 0) end up?

- A. 1
- B. 2
- C. 3**
- D. 4
- E. 5

Suppose you have ponies behind 6 doors numbered 0 1 2 3 4 5



0

1

2

3

4

5

You want to shift each pony 3 spots to the right
Where would Pinkie Pie (door 4) end up?

- A. 5
- B. 0
- C. 1
- D. 2
- E. 3

Suppose you have ponies behind 6 doors numbered 0 1 2 3 4 5



0

1

2

3

4

5

You want to shift each pony 3 spots to the right
Where would Pinkie Pie (door 4) end up?

A. 5

B. 0

C. 1

D. 2

E. 3

I think there are errors in the next slide

Let's skip the rest of these slides until we can go fix them

In general: door D, shift N spots right, T total doors

A. $D + N \% T$

B. $(D + N) \% T$

C. $(D + N) \% (N \% T)$

D. $D \% T + N$

E. not exactly one of the above

In general: door D, shift N spots right, T total doors

A. $D + N \% T$

B. $(D + N) \% T$

C. $(D + N) \% (N \% T)$

D. $D \% T + N$

E. not exactly one of the above

What about shifting left?



0

1

2

3

4

5

Discuss, and come up with a formula for shifting left.

What about shifting left? door D, shift left N, T total doors



0

1

2

3

4

5

Discuss, and come up with a formula for
shifting left.

$$(D + (T - N)) \% T$$

Let's do some examples